



Strong gravitational lensing by static black holes in effective quantum gravity

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Abstract We investigate strong gravitational lensing by two static black hole models (Model-1 and Model-2) within the Effective Quantum Gravity (EQG) framework, characterized by mass M and parameter ζ . For $\zeta = 0$, they reduce to the Schwarzschild solution, and depending on the parameters, they describe black holes with an event and Cauchy horizon (Model-1), a single horizon (Model-2), or no horizons. Using supermassive black holes (SMBHs), Sgr A* and M87*, as lenses and integrating theoretical predictions with recent Event Horizon Telescope (EHT) data, we identify significant differences in lensing signatures due to quantum corrections. For Model-1, the deviations of the lensing observables $|\delta\theta_\infty|$ of black holes in EQG from Schwarzschild black hole, for SMBHs Sgr A* and M87*, can reach as much as $1.75 \mu\text{as}$ and $1.32 \mu\text{as}$, while $|\delta s|$ is about 30.12 nas for Sgr A* and 22.63 nas for M87*. The flux ratio of the first image to all subsequent packed images indicates that EQG black hole images are brighter than their Schwarzschild counterparts, with a deviation in the brightness ratio $|\delta r_{\text{mag}}|$ reaching up to 2.02. The time delays between the second and first images, denoted $|\delta T_{2,1}|$, exhibit substantial deviations from the GR counterpart, reaching up to 1.53 min for Sgr A* and 1159.9 min for M87*. The Event Horizon Telescope (EHT) constraint on θ_{sh} of Sgr A* and M87*, within the $1\text{-}\sigma$ region, limits the parameter ζ . Our analysis concludes that EQG black holes are consistent with the EHT observations within a finite parameter space.

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1 Introduction

Gravitational lensing, a notable prediction of Einstein's General Relativity (GR), has evolved into one of the most effective tools for studying astrophysical phenomena and testing gravitational theories. Since Einstein first considered light bending around massive objects [1], the field has developed significantly, leading to groundbreaking discoveries in cosmology [2] and astrophysics [3]. The deflection of light by a gravitational field, known as gravitational lensing, plays a critical role in disclosing the nature of spacetime and the distribution of mass in the universe.

Black holes, among GR's most fascinating predictions, provide a natural arena for studying gravity in its strongest form. Developed by Einstein in 1915, GR has successfully described a wide range of astrophysical phenomena and passed numerous observational tests, from the solar system to cosmology [4]. Among these predictions, black hole regions of spacetime with gravitational fields so strong that not even light can escape have garnered considerable interest. Non-rotating black holes are represented by the Schwarzschild metric [5], while rotating black holes are modeled by the Kerr solution [6].

Despite its success, Einstein admitted that GR is incomplete at quantum scales, where it forecasts singularities at black hole centres and during the Big Bang. These issues and GR's inability to reconcile with quantum mechanics motivate the search for a more comprehensive theory of gravity, particularly in strong-field regimes. To address this, several modified theories of gravity (MTG), such as Loop Quantum Gravity (LQG) and string theory, have been proposed [7–10]. These theories offer quantum corrections to classical black hole solutions, particularly near the event horizon, result-

ing in modified geometries where quantum effects become prominent [11–16]. Such corrections could smooth out singularities and modify the behaviour of spacetime in the vicinity of black holes.

The Event Horizon Telescope (EHT) has recently provided us with the first-ever images of supermassive black holes, M87* [17] and Sgr A* [18], demonstrating the potential of direct observations to test the geometry of black holes and the validity of GR. These images reveal the shadows of black holes—corresponding to photon orbits trapped under the intense gravitational field—and have shown consistency with the Kerr black hole solution predicted by GR [19,20]. However, deviations from Kerr geometry, such as those indicated by quantum gravity, could alter the black hole shadow's size and shape and influence the associated gravitational lensing effects [21–29].

Strong gravitational lensing, where light passes near a black hole, offers precise tests of spacetime geometry. The study of strong-gravitational lensing around black holes, particularly in the presence of quantum corrections, provides a unique window into the fundamental nature of gravity. In the classical regime, the bending of light near the photon sphere has been well studied using methods such as the strong deflection limit [30], which provides analytical solutions for the deflection angle near black holes. Darwin [31] first investigated the bending of light near a Schwarzschild black hole, establishing the foundational framework for studying black hole lensing. Following developments by Virbhadra and Ellis [32,33] expanded these results, specifying relativistic images formed by light passing close to the black hole's event horizon. Frittelli et al. [34] advanced analytical methods for solving the lens equation, which Bozza later enhanced with the strong deflection limit (SDL) framework [30,35,36]. This technique has been applied to different spacetimes, such as Reissner–Nordström [37] and rotating black holes [30,38]. It has facilitated a deeper understanding of gravitational lensing near black holes. The SDL has also been instrumental in studying black holes in modified gravity models [36,39–46]. Recent research has focused on lensing in higher-curvature gravity theories [47–50], as well as modifications to Schwarzschild spacetimes [51–57], revealing new phenomena and offering crucial insights into quantum gravity.

In modified gravity scenarios, quantum corrections to black hole spacetimes can affect lensing observables like deflection angles, photon spheres, and relativistic images. Recent studies in higher-curvature gravity, LQG, and string theory [21,48,58–65] show that quantum effects alter the geometry near the event horizon, influencing black hole lensing properties. For instance, quantum corrections can shrink the lensing ring and change the photon orbits, leading to deviations in the black hole shadow that can be distinguished from classical GR solutions [60,66–68]. Comparing lens-

ing predictions in quantum-corrected spacetimes with high-resolution EHT observations provides a method to test deviations from GR and constrain quantum gravity. [18,69].

The Effective Quantum Gravity (EQG) represents one approach, incorporating quantum corrections into classical black hole solutions. Zhang et al. [70], within the Hamiltonian constraint approach to Effective Quantum Gravity (EQG), introduced two types of static black hole models that preserve general covariance, demonstrating that quantum corrections can significantly alter black hole geometry, particularly near the event horizon. Liu et al. [71] further studied the light rings and shadows of these static black holes in EQG, demonstrating that while quantum corrections do not directly affect light rings, they influence the size and characteristics of black hole shadows. It emphasizes the potential of quantum-corrected black holes to exhibit new observational signatures. The quasinormal modes and shadows of EQG black holes are studied in [72]. The gravitational lensing effects of EQG black holes in both the weak and strong field limits are investigated in [73]. They find that the quantum parameter is analogous to the electric charge in the weak gravitational lensing. In contrast, in the strong field limit, the effect of the quantum parameter is opposite to that of charges in some other gravity theories. In this work, we focus on spherically symmetric black holes in EQG [70], investigating the impact of quantum corrections on lensing observables. We aim to explore how these corrections influence the gravitational lensing signature and discuss the prospects of testing these predictions with observations like those provided by the (EHT) observations. We analyze these effects using supermassive black holes like Sgr A* and M87* as lenses. While the EHT may detect strong deflection from EQG black holes, distinguishing them from Schwarzschild black holes is challenging due to $\mathcal{O}(\mu\text{as})$ deviations.

This paper is organized as follows: Sect. 2 presents the framework of EQG, introducing two static spherically symmetric black holes and briefly discussing the horizon structure. Section 3 outlines the formalism of strong gravitational lensing in the context of EQG black holes, providing a detailed analysis of the quantum corrections on lensing observables, including deflection angles, shadows, and the time delay between the first and second images on the same side of the source. In Sect. 4, we numerically analyze the strong lensing observables by taking the supermassive black holes Sgr A* and M87* as the lens. In Sect. 5, we discuss the observational prospects for detecting these quantum corrections and how current and future experiments, such as the EHT, can be used to constrain EQG models. Finally, Sect. 6 summarizes our findings and proposes directions for future research.

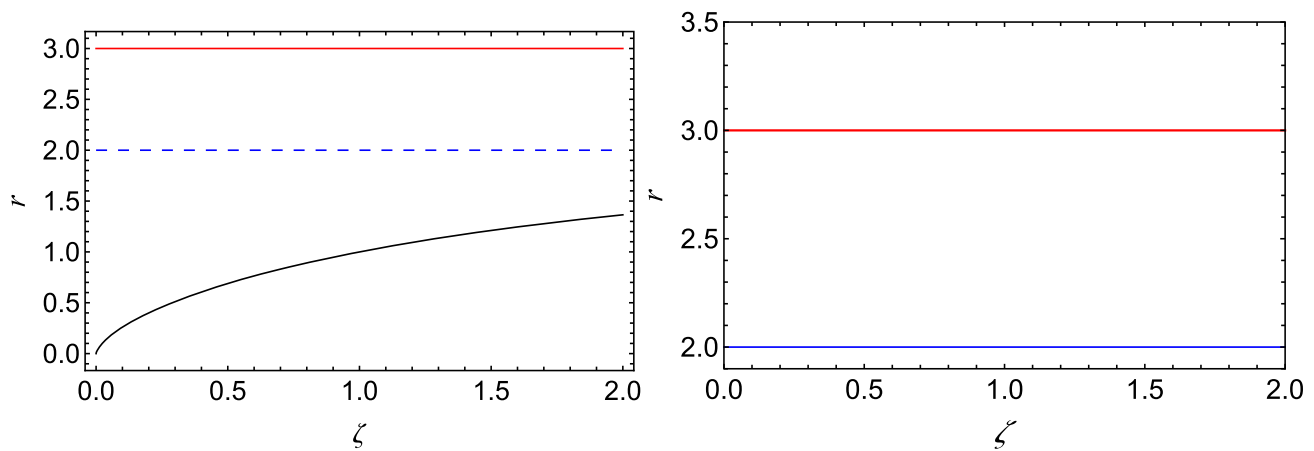


Fig. 1 Variation of horizons with parameter ζ . The blue dashed line represents the event horizon (r_+), while the solid black line represents the Cauchy horizon (r_-) for the EQG Model-1 black holes (left). The

blue solid line represents the degenerate event horizon (r_+) for EQG Model-2 black holes, and the solid red line represents the unstable photon sphere (r_m) in both cases (right)

2 Black hole metrics in EQG

Zhang et al. [70] derive stationary effective metrics in vacuum gravity while preserving covariance within the spherically symmetric sector. By retaining the kinematic variables of the theory and the classical vector constraints, they introduce an arbitrary effective Hamiltonian constraint H_{eff} and a free function to fix the gauge related to the diffeomorphism constraint. Assuming conditions similar to classical theory, including a Dirac observable for black hole mass, they establish relationships between the effective Hamiltonian, the Dirac observable, and the free function. Solving the resulting equations yields two families of quantum-corrected metrics, reducing to classical forms when the quantum parameters are zero. The metrics of the static black hole in EQG [70] are given by:

$$ds_1^2 = -A_1 dt^2 + \frac{dr^2}{B_1} + r^2 d\Omega^2, \quad (1)$$

$$A_1 = B_1 = 1 - \frac{2M}{r} + \frac{\zeta^2}{r^2} \left(1 - \frac{2M}{r}\right)^2,$$

and

$$ds_2^2 = -A_2 dt^2 + \frac{dr^2}{B_2} + r^2 d\Omega^2, \quad (2)$$

$$A_2 = 1 - \frac{2M}{r}, \quad B_2 = A_2 \left(1 + A_2 \frac{\zeta^2}{r^2}\right),$$

with $d\Omega^2 = d\theta^2 + \sin^2\theta d\phi^2$, and ζ is the quantum parameter, while M denotes the ADM mass. These metrics will hereafter be referred to as Model-1 and Model-2. The metric (1) has two positive roots for all $M > 0$: $r_+ = 2M$ and $r_- = \zeta^2/\alpha - \alpha/3$, with $\alpha^3 = 3\zeta^2(\sqrt{81M^2 + 3\zeta^2} - 9M)$. These roots correspond to the event horizon (r_+) and the Cauchy horizon (r_-), respectively, as depicted in Fig. 1. This

indicates that spacetime (1) exhibits a double-horizon structure, similar to a Reissner–Nordström black hole with a small charge. For the metric (2), we get a degenerate horizon at $r = 2M$.

Henceforth, we will perform strong gravitational lensing calculations for the general metric

$$ds_1^2 = -A(r)dt^2 + \frac{dr^2}{B(r)} + r^2 d\Omega^2, \quad (3)$$

and carry out numerical evaluations for Model-1 using $A = A_1$ and $B = B_1$, and similarly for Model-2 with $A = A_2$ and $B = B_2$.

3 Strong Gravitational Lensing by EQG Black Holes

This section discusses strong gravitational lensing by EQG black holes and analyzes how the quantum parameter ζ affects the deflection angle and lensing observables. We begin by examining the propagation of light in the equatorial plane ($\theta = \frac{\pi}{2}$), given the spherical symmetry. The conservation of photon four-momentum along the Killing vectors of isometry yields two invariant quantities: energy, $\mathcal{E}$, defined as,

$$\mathcal{E} = -p_\mu \xi_{(t)}^\mu,$$

and angular momentum $\mathcal{L}$, defined as

$$\mathcal{L} = p_\mu \xi_{(\phi)}^\mu,$$

where $\xi_{(t)}^\mu$ and $\xi_{(\phi)}^\mu$ are the Killing vectors associated with time and rotational invariance, respectively. Since photons follow null geodesics with $ds^2 = 0$, this null condition leads to

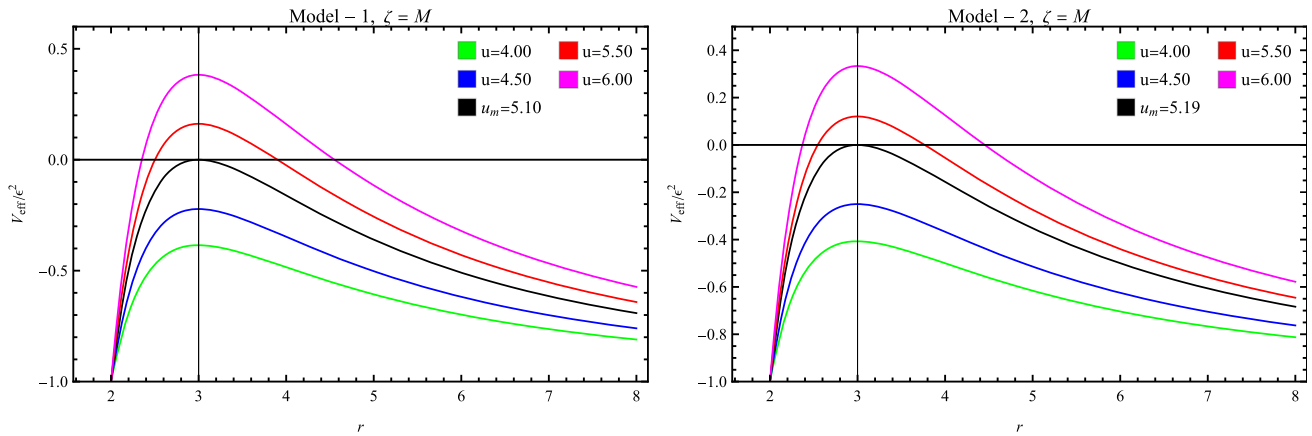


Fig. 2 Effective potential of photon for Model-1 and Model-2 with the quantum parameter $\zeta = M$ at marginally bound orbit. The brown curves denote the unstable circular orbit that $u = u_m$, on which the pho-

ton can sometimes encircle around the black hole. The black hole will capture photons with $u < u_m$, while the larger ones will be scattered

$$\frac{A(r)}{B(r)} \left(\frac{dr}{d\tau} \right)^2 + \frac{A(r)}{r^2} \mathcal{L}^2 = \mathcal{E}^2. \quad (4)$$

The effective potential, in terms of impact parameter u , can be written as

$$V_{\text{eff}}(r)/\mathcal{E}^2 = \frac{A(r)u^2}{r^2} - 1, \quad (5)$$

where u is defined by $u = \mathcal{L}/\mathcal{E}$. Figure 2 shows the effective potential for Model-1 and Model-2.

The photon sphere exists when $V_{\text{eff}}(r)/\mathcal{E}^2 \leq 0$. Meanwhile, the unstable circular orbit is defined by the conditions $V_{\text{eff}}(r) = V'_{\text{eff}}(r) = 0$ and $V''_{\text{eff}}(r) < 0$. The largest positive root that satisfies these conditions is the radius of the unstable photon sphere r_m . Solving these conditions yields $r_m = 3M$, corresponding to an impact parameter $u = u_m$, which is consistent with the Schwarzschild black hole value and independent of the parameter ζ .

Throughout the trajectory that the photon moves straightforwardly to the black hole and then is redirected to the observer, the impact parameter u , which represents the perpendicular distance from the black hole's center to the photon's initial direction at infinity, remains constant due to spacetime symmetry. It can be represented as a function of the minimum approach distance r_0 [74], where the effective potential vanishes, i.e., $V_{\text{eff}}(r_0) = 0$. Thus, it can be expressed as

$$u = \frac{\mathcal{L}}{\mathcal{E}} = \sqrt{\frac{r_0^2}{A(r_0)}}. \quad (6)$$

At $r_0 = r_m$, the impact parameter is called critical impact parameter u_m shown in Fig 3. The deflection angle provides significant information about gravitational lensing and can be extracted by solving the null geodesic equation for a photon

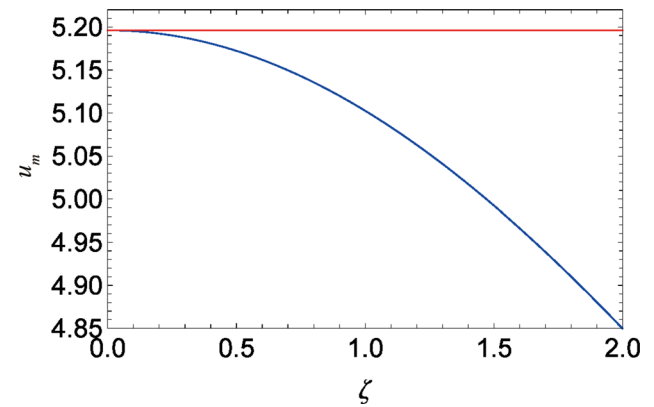


Fig. 3 Variation of critical impact parameter u_m with the parameter ζ for EQG black holes. Note that the critical impact parameters for Model-1 and Model-2 ($\approx$ Schwarzschild black hole), are represented by blue and red lines, respectively

on the equatorial plane:

$$\frac{d\phi}{dr} = \sqrt{\frac{Ar_0^2}{Br^2(A_0r^2 - Ar_0^2)}}, \quad (7)$$

where the functions with subscript 0 are evaluated at r_0 . Then the deflection angle reads as

$$I(r_0) = 2 \int_{r_0}^{\infty} \frac{d\phi}{dr} dr, \quad (8)$$

$$\alpha_D(r_0) = I(r_0) - \pi. \quad (9)$$

For all spherically symmetric metrics, the deflection angle should be logarithmic [75], which can be rewritten in terms of impact parameter u as

$$\alpha_D(u) = -\bar{a} \log \left(\frac{u}{u_m} - 1 \right) + \bar{b} + \mathcal{O}(u - u_m), \quad (10)$$

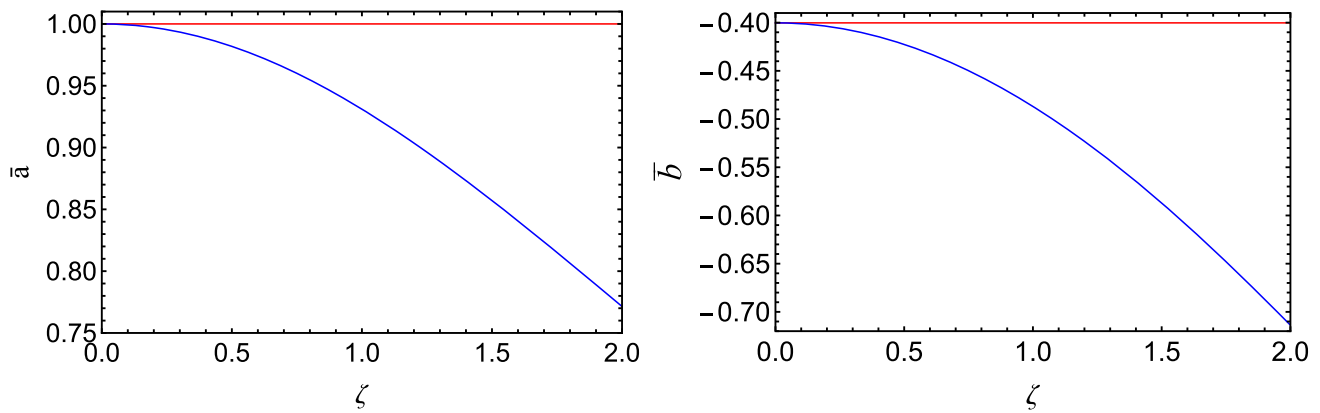


Fig. 4 Behavior of strong lensing coefficients $\bar{a}$ and $\bar{b}$ along with the parameter ζ . Note that the lensing coefficients for Model-1 and Model-2 are represented by blue and red lines, respectively

with $u_m = u(r_0 = r_m)$, and $\bar{a}$ and $\bar{b}$ denoting the lensing coefficients. To obtain these two coefficients, we define z as $z = \frac{A-A_0}{1-A_0}$, then the integral equals to

$$I(r_0) = \int_0^1 R(z, r_0) F(z, r_0) dz, \quad (11)$$

where

$$R(z, r_0) = \frac{2(1-A_0)\sqrt{Ar_0^2}}{r^2 A' \sqrt{B}}, \quad (12)$$

$$F(z, r_0) = \frac{r}{\sqrt{A_0 r^2 - A r_0^2}}. \quad (13)$$

The function $R(z, r_0)$ is regular for all z , but $F(z, r_0)$ will diverge as $z \rightarrow 0$. To avoid the divergence, we expand the square root term of $F(z, r_0)$ to the second order in z in the Taylor series

$$F(z, r_0) \sim \bar{F}(z, r_0) = \frac{1}{\sqrt{c_0 z + d_0 z^2}}. \quad (14)$$

See Appendix A for the detailed calculations. To solve Eq. (11), it will be split into two parts: I_D acts as a divergent part, and I_R is regular by subtracting the divergence from the original integral. In other word, $I(r_0) = I_D(r_0) + I_R(r_0)$ with

$$I_D(r_0) = \int_0^1 R(0, r_m) \bar{F}(z, r_0) dz, \quad (15)$$

$$I_R(r_0) = \int_0^1 [R(z, r_0) F(z, r_0) - R(0, r_m) \bar{F}(z, r_0)] dz. \quad (16)$$

The lensing coefficients $\bar{a}$ and $\bar{b}$ are the result of the above two integrals, which can be described as

$$\bar{a} = \frac{R(0, r_m)}{2\sqrt{d_m}}, \quad (17)$$

$$\bar{b} = -\pi + I_R(r_m) + \bar{a} \log\left(\frac{2d_m}{A_m}\right), \quad (18)$$

with

$$d_m = d_0|_{r_0=r_m} = \frac{r_m^2 (1-A_m)^2 (2A_m - A_m'' r_m^2)}{4A_m^2 r_m^2}. \quad (19)$$

Refer to Appendix A for comprehensive calculations. We use numerical solutions to study the relationship between the strong gravitational lensing coefficients and the quantum parameter. As shown in Fig. 4, it is evident that both the deflection coefficients $\bar{a}$ and $\bar{b}$ gradually decrease as the quantum parameter ζ increases for Model-1. It is important to note that when the quantum parameter $\zeta = 0$, the EQG black holes for Model-1 and Model-2 reduce to the Schwarzschild black hole, and our calculations yield the values [75]: $\bar{a} = 1$ and $\bar{b} = -0.40023$ (see Table 1). Figure 5 shows the typical deflection angles. As u_0 approaches u_m , the deflection angle increases logarithmically and exceeds 2π , indicating that the photon, close to the critical, unstable circular orbit, encircles the black hole multiple times before being redirected to the observer. As the photon is away from the u_m , the deflection angle becomes negative, which shows that the photon is reflected opposite the black hole. But Eq. (10) applies to strong gravitational lensing for the vicinity of the critical impact parameter; The deflection angle at the larger impact parameter is biased and must be obtained using the weak limit method.

3.1 Lens equation and observables

Assuming that the observer and the light source are both far from the black hole, the asymptotically flat lens equation is derived as [35]

$$\psi = \theta - \frac{D_{\text{LS}}}{D_{\text{OS}}} \Delta\alpha_n, \quad (20)$$

Table 1 Estimates for the strong lensing coefficients $\bar{a}$, $\bar{b}$ and the critical impact parameter u_m/R_s for the EQG black. Here, the value $\zeta = 0$ is the Schwarzschild black hole equivalent to the EQG Model-2 black hole. Note that, u_m is in the units of Schwarzschild radius $R_s = 2GM/c^2$

ζ	$\bar{a}$	$\bar{b}$	u_m/R_s
0.0	1.	−0.400237	5.19615
0.5M	0.981818	−0.422501	5.17226
1.0M	0.931034	−0.486982	5.10252
1.5M	0.857143	−0.587184	4.9923
2.0M	0.771429	−0.713767	4.84934

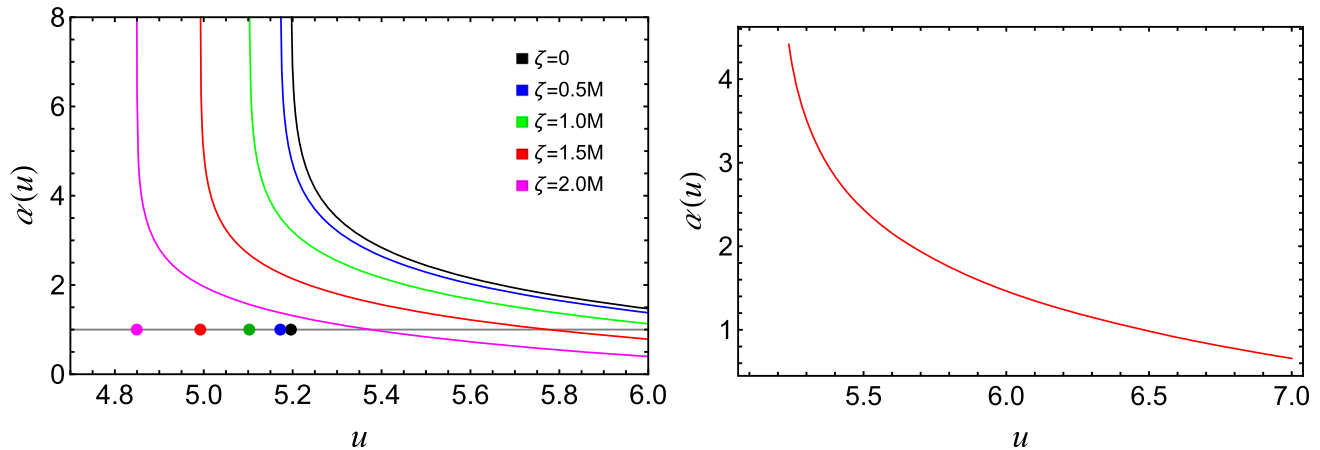


Fig. 5 (Left) Deflection angle as a function of the impact parameter u for different quantum parameters ζ for EQG Model-1 black holes. The points on the horizontal axis denote the values of the impact parameter

$u = u_m$ at which the deflection angle diverges. The black curves denote the Schwarzschild black hole. (Right) Deflection angle as a parameter ζ function for EQG Model-2 black holes

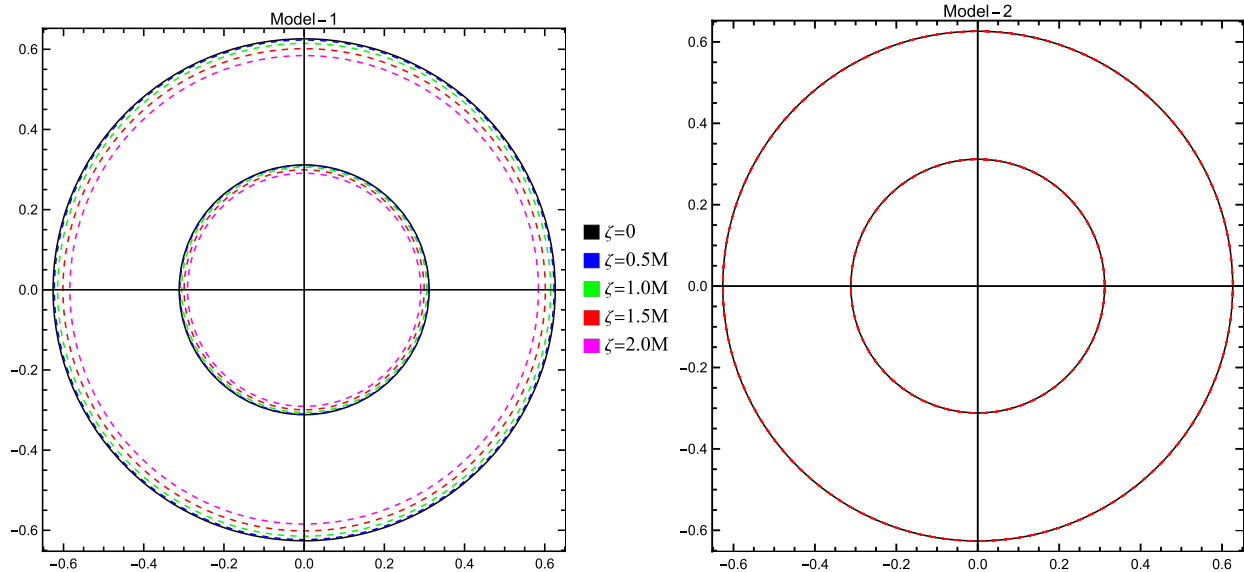
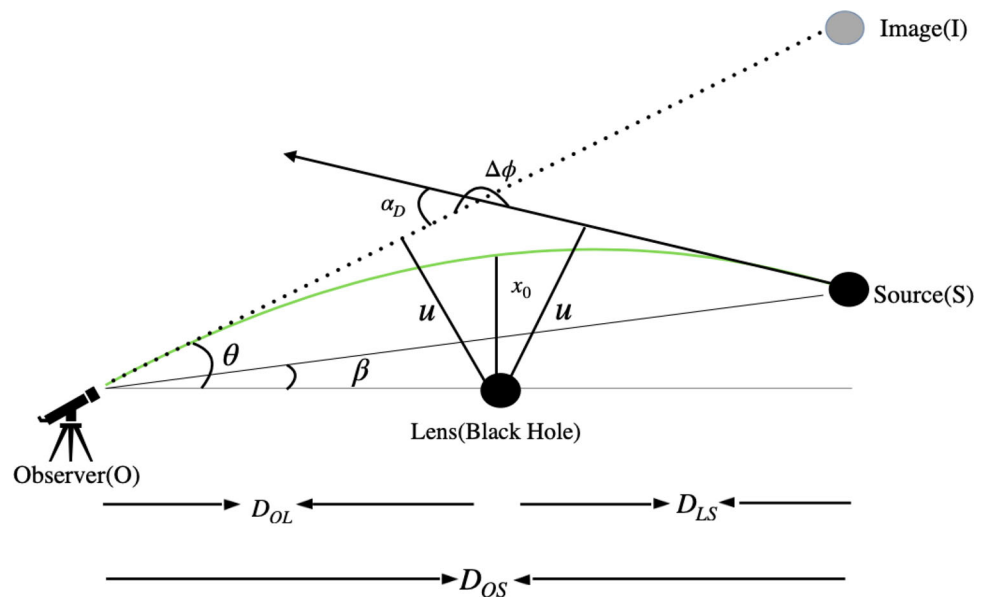


Fig. 6 The outermost relativistic Einstein ring about Sgr A* (outer $D_{OL} = 9.3$ kpc) and M87* (inner $D_{OL} = 16.68$ Mpc) for Model-1 and Model-2 with different quantum parameters ζ . Black curves denote

the Schwarzschild black hole. For Model-2, the radius is independent of the quantum parameter, ζ that all rings overlap into one

Fig. 7 A schematic diagram illustrating the configuration of gravitational lensing, where the observer, lens, light source, and image are labeled as O, L, S, and I, respectively



where the subscripts S , L , and O refer to the source, lens, and observer, respectively (cf. Fig. 7), D denotes the distance, and θ , and ψ are the angular positions of the image and source. The $\Delta\alpha_n = \alpha_D(\theta) - 2n\pi$ is the offset of deflection angle, where $u = \theta D_{OL}$, and n counts the circles. Using Eqs. (10) and (20), and the relation $u \approx \theta D_{OL}$, the angular position of n^{th} relativistic image in terms of the strong lensing coefficients is given as [36]

$$\theta_n = \theta_n^0 + \frac{u_m e_n (\psi - \theta_n^0) D_{OS}}{\bar{a} D_{LS} D_{OL}}, \quad (21)$$

where

$$e_n = \exp\left(\frac{\bar{b} - 2n\pi}{\bar{a}}\right), \quad (22)$$

$$\theta_n^0 = \frac{u_m (1 + e_n)}{D_{OL}}. \quad (23)$$

Thus, the magnification of n th relativistic image is given by [36]

$$\mu_n = \left(\frac{\psi}{\theta} \frac{d\psi}{d\theta}\right)^{-1} \bigg|_{\theta_n^0} = \frac{u_m^2 e_n (1 + e_n) D_{OS}}{\bar{a} \psi D_{LS} D_{OL}^2}, \quad (24)$$

which decreases exponentially with n and is inversely proportional to D_{OL}^2 . When the black hole, source, and observer are aligned, with the black hole at the center, i.e. $\psi = 0$ and $D_{OS} = D_{LS} = 2D_{OL}$, the Einstein ring radius yields

$$\theta_n^E = \left(1 - \frac{u_m e_n}{\bar{a} D_{OL}}\right) \theta_n^0. \quad (25)$$

Following [76] and EHT [77], we estimate the outermost Einstein ring ($n = 1$), for Sgr A* with $D_{OL} = 7.97$ kpc and for M87* with $D_{OL} = 16.8$ Mpc. As ζ increases, the radius

of both rings decreases (cf. Fig. 6). For higher orders, the Einstein ring shrinks slightly.

Apart from the Einstein ring, there are three other essential observables: θ_∞ , which represents the asymptotic position approached by a set of images; s , which describes the separation between θ_∞ and θ_1 ; and r_{mag} , which is the ratio of the flux of the first image to the combined flux of all other images. These observables are given by [78]

$$\theta_\infty = \frac{u_m}{D_{OL}}, \quad (26)$$

$$s = \theta_1 - \theta_\infty \approx \theta_\infty \exp\left(\frac{\bar{b} - 2\pi}{\bar{a}}\right), \quad (27)$$

$$r_{\text{mag}} = \frac{\mu_1}{\sum_{n=2}^{\infty} \mu_n} \approx \frac{5\pi}{\bar{a} \log(10)}. \quad (28)$$

From their astronomical observations, we can easily reconstruct the lensing parameters $\bar{a}$ and $\bar{b}$ and then examine the equation (Table 2).

3.2 Time delay

Since photons take different amounts of time to encircle the black hole and travel to various images, a time delay occurs between the images. The total time taken by the photon to travel from the source to the observer is given by [79, 80]

$$T(r_0) = \tilde{T}(r_0) - \int_{D_{OL}}^{\infty} \left| \frac{dt}{dr} \right| dr - \int_{D_{LS}}^{\infty} \left| \frac{dt}{dr} \right| dr, \quad (29)$$

where

$$\tilde{T}(r_0) = \int_{r_0}^{\infty} \frac{2rr_0\sqrt{A_0}}{A\sqrt{A_0r^2 - Ar_0^2}} dr. \quad (30)$$

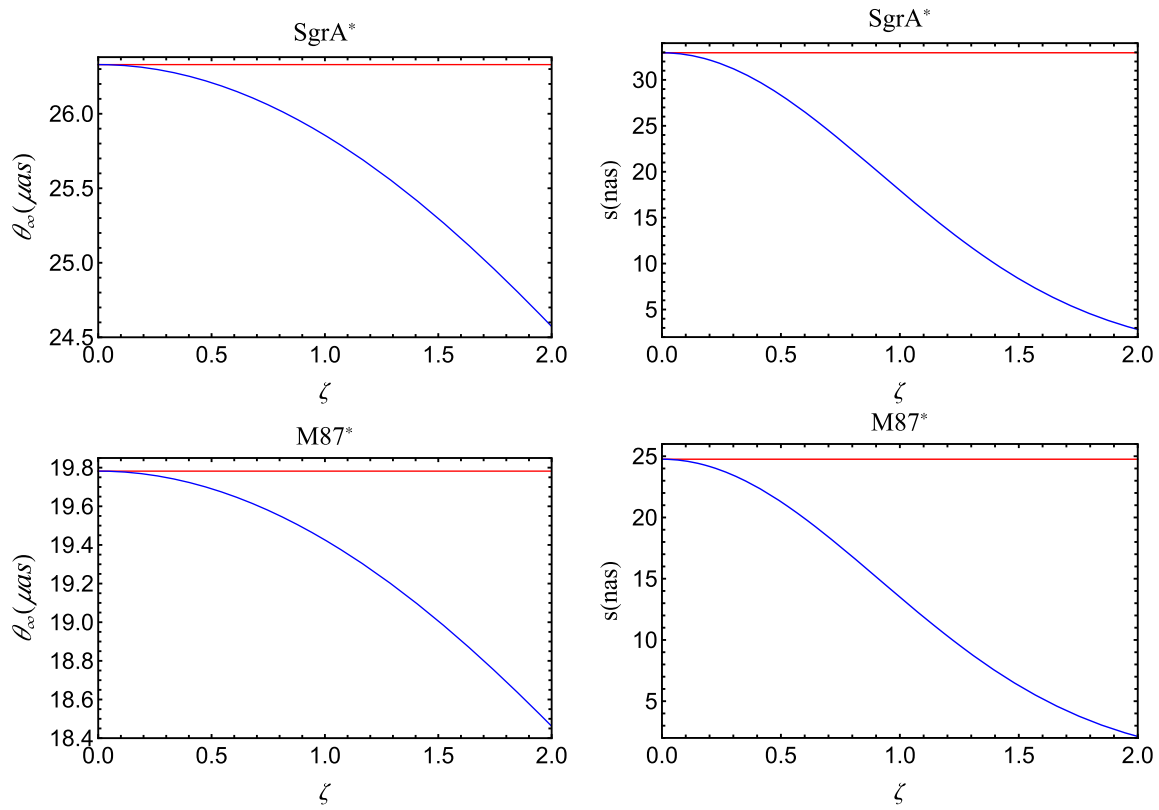


Fig. 8 Behavior of Strong Lensing Observables θ_∞ and s with the parameter ζ by taking supermassive black holes Sgr A* (Upper panel) and M87* (Lower panel) as the static black hole effective quantum

gravity. Note that the lensing observables for Model-1 and Model-2 are represented by blue and red lines, respectively

Table 2 The estimation of lensing observables for Sgr A* ($M = 4.3 \times 10^6 M_\odot$, $D_{\text{OL}} = 8.3 \text{ kpc}$) and M87* ($M = 6.15 \times 10^9 M_\odot$, $D_{\text{OL}} = 16.68 \text{ Mpc}$) considered as EQG black holes for Model-1. Here,

the value $\zeta = 0$ corresponds to the Schwarzschild black hole, equivalent to the EQG Model-2 black hole

ζ	Sgr A*		M87*		r_{mag}
	θ_∞ (μas)	s (nas)	θ_∞ (μas)	s (nas)	
0.0	26.3299	32.9515	19.782	24.757	6.82188
0.5M	26.2088	28.3318	19.6911	21.2861	6.94821
1.0M	25.8554	17.9684	19.4256	13.4999	7.32721
1.5M	25.2969	8.35623	19.006	6.27816	7.95886
2.0M	24.5725	2.82713	18.4617	2.12406	8.84318

The integral $\tilde{T}(r_0)$ describes the time taken by the photon to encircle the black hole. Under the assumption that the source and observer are far from the black hole, the last two integrals cancel each other. Similar to deflection angle, $\tilde{T}(r_0)$ should have a logarithmic form:

$$\tilde{T}(u) = -\tilde{a} \log\left(\frac{u}{u_m} - 1\right) + \tilde{b} + \mathcal{O}(u - u_m), \quad (31)$$

where $\tilde{a}$ and $\tilde{b}$ are strong deflection limit coefficients [79].

Assuming the source is variable and the difference between the signals from the first and second images can be distinguished, then [79]

$$\Delta T_{2,1}^s = 2\pi u_m = 2\pi \theta_\infty D_{\text{OL}}. \quad (32)$$

By measuring the time delay, we can estimate the distance of the black hole.

4 Lensing effects of the supermassive black holes Sgr A* and M87*

In this section, we estimate and compare the lensing observables θ_∞ , s , and r_{mag} for EQG black holes with those of Schwarzschild black holes, using the supermassive black holes M87* and Sgr A* as potential EQG black holes can-

Table 3 Estimation and comparison of time delay for supermassive black holes at the centre of nearby galaxies between the EQG Model-1 ($\zeta = 2M$), EQG Model-2, and Schwarzschild black hole ($\zeta = 0$). Themass (M) and the distance (D_{OL}) are given in units of solar mass and Mpc, respectively. Time delays are expressed in minutes

Galaxy	$M(M_{\odot})$	D_{OL} (Mpc)	M/D_{OL}	$\Delta T_{2,1}^s$ (GR)	$\Delta T_{2,1}^s$ (Model-1)	$\Delta T_{2,1}^s$ (Model-2)
Milky Way	4.3×10^6	0.0083	2.471×10^{-11}	11.4968	10.7294	11.4968
M87*	6.5×10^9	16.68	1.758×10^{-11}	17378.8	16218.91	17378.8
NGC 4472	2.54×10^9	16.72	7.246×10^{-12}	6791.11	6337.85	6791.11
NGC 1332	1.47×10^9	22.66	3.094×10^{-12}	3930.29	3667.97	3930.29
NGC 4374	9.25×10^8	18.51	2.383×10^{-12}	2473.14	2308.08	2473.14
NGC 1399	8.81×10^8	20.85	2.015×10^{-12}	2355.5	2198.29	2355.5
NGC 3379	4.16×10^8	10.70	1.854×10^{-12}	1112.25	1038.01	1112.25
NGC 4486B	6×10^8	16.26	1.760×10^{-12}	1604.2	1497.13	1604.2
NGC 1374	5.90×10^8	19.57	1.438×10^{-12}	1577.46	1472.18	1577.46
NGC 4649	4.72×10^9	16.46	1.367×10^{-12}	12619.7	11777.4	12619.7
NGC 3608	4.65×10^8	22.75	9.750×10^{-13}	1243.26	1160.28	1243.26
NGC 3377	1.78×10^8	10.99	7.726×10^{-13}	475.913	444.149	475.913
NGC 4697	2.02×10^8	12.54	7.684×10^{-13}	540.081	504.034	540.081
NGC 5128	5.69×10^7	3.62	7.498×10^{-13}	152.132	141.978	152.132
NGC 1316	1.69×10^8	20.95	3.848×10^{-13}	451.85	421.69	451.85
NGC 3607	1.37×10^8	22.65	2.885×10^{-13}	366.292	341.845	366.292
NGC 4473	0.90×10^8	15.25	2.815×10^{-13}	240.63	224.57	240.63
NGC 4459	6.96×10^7	16.01	2.073×10^{-13}	186.087	173.667	186.087
M32	2.45×10^6	0.8057	1.450×10^{-13}	6.55048	6.11328	6.55048
NGC 4486A	1.44×10^7	18.36	3.741×10^{-14}	38.5008	35.93	38.5008
NGC 4382	1.30×10^7	17.88	3.468×10^{-14}	34.7577	32.4378	34.7577
CYGNUS A	2.66×10^9	242.7	1.4174×10^{-15}	7111.95	6637.28	7111.95

didates. Based on the latest astronomical observations, the mass and distance from Earth for M87* are approximately $(6.5 \pm 0.7) \times 10^9 M_{\odot}$ and 16.8 Mpc [69, 81] respectively. For Sgr A* they are about $4.0_{-0.6}^{+1.1} \times 10^6 M_{\odot}$ and 7.97 kpc respectively. [82, 83].

Using Eq. (21) and assuming $d = D_{\text{LS}}/D_{\text{OS}} = 1/2$, we calculated the angular positions of the first- and second-order relativistic primary and secondary images for Sgr A* and M87*. Our results show that, in EQG black holes, the angular positions of the images are smaller than in the Schwarzschild black hole and exhibit minimal sensitivity to the source position ψ . For larger values of ψ , $\theta_{1p} > |\theta_{1s}|$, while for smaller ψ , the values converge closely. This behaviour is consistent with higher-order relativistic images. In EQG, at $\zeta = 2M$, the first and second-order primary images are approximately $1.78 \mu\text{as}$ and $1.75 \mu\text{as}$ larger than their GR counterparts. However, this difference is too small to be detected with current telescopes, particularly due to the strong demagnification of these relativistic images.

Comparing the strong lensing observations for Model-2 of EQG black holes with those for Model-1, which behaves similarly to the Schwarzschild black hole case ($\zeta = 0$),

we observe that the angular separation s between images is smaller. At the same time, their relative magnification r_{mag} is larger than in the Schwarzschild case. For Sgr A*, the angular position θ_{∞} ranges from 24.57 to 26.33 μas , while for M87* it ranges from 18.46 to 19.78 μas . The angular separation s for Sgr A* ranges from 2.8327 to 32.95 nas and for M87* from 2.12 to 24.75 nas. The deviations of the lensing observables $|\delta s|$ and $|\delta \theta_{\infty}|$ for EQG black holes from the Schwarzschild black hole can reach up to 30.12 nas and $1.75 \mu\text{as}$ for Sgr A*, and 22.63 nas and $1.32 \mu\text{as}$ for M87*, respectively. Furthermore, for fixed parameter values, the angular separation between relativistic images rendered by the Sgr A* black hole exceeds that of the M87* black hole. Therefore, EQG black hole images are brighter than Schwarzschild's black hole images, a phenomenon apparent from the increasing flux ratio of the first image relative to all other images as ζ increases. Using Eq. (24), the relative magnification of the first- and second-order images are tabulated in Tables 7 and 8 for black holes in GR and EQG at $\zeta = 2M$. The first-order images in EQG are significantly more magnified than the second-order images and their corresponding images in GR. Compared to Schwarzschild black hole

Table 4 The deviation of the Schwarzschild black hole from the EQG black holes of Model-1 for different values of the parameter ζ has been calculated where $\delta\theta_\infty = \theta_\infty(EQG) - \theta_\infty(Sch)$ and $\delta s(nas) = s_{EQG} - s_{Sch}$. Note that for EQG Model-2 $\delta\theta_\infty$ and $\delta s(nas)$ is 0

ζ	Sgr A*		M87*		δr_{mag}
	$\delta\theta_\infty(\mu as)$	$\delta s(nas)$	$\delta\theta_\infty(\mu as)$	$\delta s(nas)$	
0.5M	-0.1211	-4.6197	-0.091	-3.4709	0.1263
1.0M	-0.4745	-14.9831	-0.3564	-11.2571	0.5053
1.5M	-1.033	-24.5953	-0.776	-18.4789	1.137
2.0M	-1.7574	-30.1244	-1.3203	-22.633	2.0213

images, the brightness of EQG black hole images increases with higher values of ζ , as shown in Fig. 8. Table 3 presents updated data on 22 supermassive black holes with varying masses and distances from Sgr A*. Using this data, we calculated the time delays between the first- and second-order relativistic primary images ($\Delta T_{2,1}^s$). For Sgr A* and M87*, the time delays can reach ~ 10.73 min and ~ 16218.9 min at $\zeta = 2M$, respectively. These values deviate from their GR counterparts ~ 0.7674 min for Sgr A* and ~ 1159.9 min for M87*. While these deviations are negligible for Sgr A*, they are significant for M87* and other black holes, providing valuable insights for testing and comparing EQG black holes with those predicted by general relativity (Tables 4, 5, 6, 7, 8).

5 EHT constraints on the observables

In 2019, the Event Horizon Telescope (EHT) Collaboration achieved a groundbreaking achievement by capturing the horizon-scale image of the supermassive black hole M87* [17, 69, 81]. With a distance of $d = 16.8$ Mpc and an estimated mass of $M = (6.5 \pm 0.7) \times 10^9 M_\odot$, the EHT Collaboration determined that M87* has a compact emission region with an angular diameter of $\theta_d = 42 \pm 3 \mu as$. The central flux depression, which indicates the shadow of the black hole, is at least a factor of 10. In 2022, the EHT observed the supermassive black hole Sgr A* at the centre of the Milky Way, revealing an image with a diameter ring $\theta_d = 48.7 \pm 7 \mu as$. This observation suggests a black hole mass of $M = 4.0_{-0.6}^{+1.1} \times 10^6 M_\odot$ and a Schwarzschild shadow deviation of $\delta = -0.08_{-0.09}^{+0.09}$ (VLTI), $-0.04_{-0.10}^{+0.09}$ (Keck). These findings offer a valuable new approach to probing strong field gravity [18, 82, 84]. The EHT also calculated an angular diameter of the emission ring for Sgr A* of $\theta_d = (51.8 \pm 2.3) \mu as$, based on the previously estimated mass M and $D_{LS} = 8.15 \pm 0.15$ kpc [18].

The horizon-scale images of these supermassive black holes provide a novel theoretical framework for testing gravity theories [13, 18, 50, 85–97]. By analyzing the EHT observations of the shadows of Sgr A* and M87*, we can constrain the deviation parameter ζ for the EQG black hole spacetime

in Model-1. Using the apparent radius of the photon sphere θ_∞ , as a proxy for the angular size of the black hole shadow, we can constrain this deviation parameter within the $1-\sigma$ confidence level.

Constraints from M87*: We find that the Schwarzschild black hole ($\zeta = 0$) casts the largest shadow, with an angular diameter of $\theta_{sh} = 2\theta_\infty = 39.56069 \mu as$. This value falls within the $1-\sigma$ region for a black hole with mass $M = (6.5 \pm 0.7) \times 10^9 M_\odot$ and distance $D_{OL} = 16.8$ Mpc [17, 69, 81]. Figure 9 shows the angular diameter θ_{sh} as a function of ζ , with the black solid line representing $\theta_{sh} = 39 \mu as$ for the EQG black hole spacetime corresponding to M87*. When the EQG black hole spacetime metric is analyzed using the EHT results for M87* within the $1-\sigma$ bound, the parameter ζ is constrained to $0 < \zeta \leq 0.9M$. Thus, based on Fig. 9, the EQG black hole spacetime could be a candidate for astrophysical black holes within only a tiny part of the parameter space.

Constraints from Sgr A*: The EHT observation employed three independent algorithms, EHT-Imaging, SMILI, and DIFMAP to determine that the mean measured value of the shadow angular diameter lies in the range $\theta_{sh} \in (46.9, 50) \mu as$, with a $1-\sigma$ interval of $(41.7, 55.6) \mu as$ [18]. The angular diameter $\theta_{sh} \in (41.7, 55.6) \mu as$ falls within the $1-\sigma$ confidence region of the EHT observations of the Sgr A* black hole shadow, constraining the parameter ζ to $0.0 \leq \zeta \leq 4M$ for the EQG black hole spacetime. Thus, within the finite parameter space, the EQG black hole spacetime is consistent with the EHT results for the Sgr A* black hole shadow (see Fig. 9).

6 Conclusion

Recent developments in the EQG framework have led to significant modifications of black hole spacetimes through quantum corrections. Zhang et al. [70] derived stationary effective metrics while preserving covariance in spherical symmetry. By retaining the kinematic variables and classical vector constraints, they formulated an arbitrary effective Hamiltonian constraint H_{eff} and a free function addressing the gauge associated with the diffeomorphism constraint.

Table 5 Image positions of first and second-order images due to lensing by Sgr A* with $d = D_{\text{LS}}/D_{\text{OS}} = 1/2$: EQG Model-2, which is equivalent to the Schwarzschild black hole, and EQG Model-1 ($\zeta = 2M$) predictions for angular positions θ of primary (p) and secondary images

ψ	EQG				GR			
	$\theta_{1p,\text{EQG}}$	$\theta_{2p,\text{EQG}}$	$\theta_{1s,\text{EQG}}$	$\theta_{2s,\text{EQG}}$	$\theta_{1p,\text{GR}}$	$\theta_{2p,\text{GR}}$	$\theta_{1s,\text{GR}}$	$\theta_{2s,\text{GR}}$
0	24.5753	24.5725	− 24.5753	− 24.5725	26.3628	26.3299	− 26.3628	− 26.3299
10^0	24.5753	24.5725	− 24.5753	− 24.5725	26.3628	26.3299	− 26.3628	− 26.3299
10^1	24.5753	24.5725	− 24.5753	− 24.5725	26.3628	26.3299	− 26.3628	− 26.3299
10^2	24.5753	24.5725	− 24.5753	− 24.5725	26.3628	26.3299	− 26.3628	− 26.3299
10^3	24.5754	24.5725	− 24.5753	− 24.5725	26.3631	26.3299	− 26.3625	− 26.3299
10^4	24.5757	24.5725	− 24.575	− 24.5725	26.366	26.3299	− 26.3596	− 26.3299

(s) are given for different values of angular source position ψ . **(a)** All angles are in μas . **(b)** We have used $M_{\text{Sgr A}^*} = 4.3 \times 10^6 M_{\odot}$, $D_{\text{OL}} = 8.3 \times 10^6 \text{ pc}$

Table 6 Image positions of the first and second-order images due to lensing by M87* with $d = D_{\text{LS}}/D_{\text{OS}} = 1/2$: EQG Model-2, which is equivalent to Schwarzschild black hole and EQG Model-1 ($\zeta = 2M$) predictions for angular positions θ of primary (p) and secondary images

ψ	EQG				GR			
	$\theta_{1p,\text{EQG}}$	$\theta_{2p,\text{EQG}}$	$\theta_{1s,\text{EQG}}$	$\theta_{2s,\text{EQG}}$	$\theta_{1p,\text{GR}}$	$\theta_{2p,\text{GR}}$	$\theta_{1s,\text{GR}}$	$\theta_{2s,\text{GR}}$
0	18.4638	18.4617	− 18.4638	− 18.4617	19.8068	19.7821	− 19.8068	− 19.7821
10^0	18.4638	18.4617	− 18.4638	− 18.4617	19.8068	19.7821	− 19.8068	− 19.7821
10^1	18.4638	18.4617	− 18.4638	− 18.4617	19.8068	19.7821	− 19.8068	− 19.7821
10^2	18.4638	18.4617	− 18.4638	− 18.4617	19.8068	19.7821	− 19.8068	− 19.7821
10^3	18.4639	18.4617	− 18.4638	− 18.4617	19.807	19.7821	− 19.8065	− 19.7821
10^4	18.4641	18.4617	− 18.4636	− 18.4617	19.8092	19.7821	− 19.8044	− 19.7821

(s) are given for different values of angular source position ψ . **(a)** All angles are in μas . **(b)** We have used $M_{\text{M87}^*} = 6.5 \times 10^9 M_{\odot}$, $D_{\text{OL}} = 16.8 \times 10^6 \text{ pc}$

This approach yielded two families of effective Hamiltonian constraints, each characterized by a quantum parameter. When these parameters vanish, the constraints reduce to their classical forms, resulting in two quantum-corrected metrics with distinct spacetime structures. Building on this work, we explored strong gravitational lensing by these black holes (Model-1 and Model-2), which generalize the Schwarzschild black hole by introducing the parameter ζ . In our study, we analyzed both Model-1 and Model-2 together. Model-2 behaves similarly to Schwarzschild black holes, limiting its novelty compared to Model-1, where quantum corrections result in distinct deviations. A comparative discussion, supported by tables and figures, has been included to highlight these findings. Our analysis shows that quantum corrections significantly influence the lensing observables for Model-1, while Model-2 remains indistinguishable from the Schwarzschild solution.

We examined the impact of ζ on lensing coefficients ($\bar{a}$, $\bar{b}$) and the deflection angle. For Model-1, the coefficients decrease with increasing ζ , while Model-2 shows no dependence on ζ . For a fixed impact parameter, the deflection angle α_D is smaller for EQG black holes than for GR ones. Using supermassive black holes M87* and Sgr A* as lenses, we found that quantum corrections lead to deviations in key lens-

ing features, such as angular positions, image brightness, and time delays. The brightness of the first image increases, while the time delays between subsequent images also increase. Our results indicate that the angular position θ_{∞} and the image separation s decrease as ζ increases, whereas the brightness ratio r_{mag} increases. The Einstein ring for EQG black holes is smaller than that for the GR black hole.

We found that the relativistic image angular position θ_{∞} varies between $18.4617 \mu\text{as} \leq \theta_{\infty}(\text{M87}^*) \leq 19.7821 \mu\text{as}$ and $24.5725 \mu\text{as} \leq \theta_{\infty}(\text{Sgr A}^*) \leq 26.3299 \mu\text{as}$. The deviation $|\delta\theta_{\infty}|$ can reach $1.75 \mu\text{as}$ for Sgr A* and $1.32 \mu\text{as}$ for M87*. The time delays between the second and first images, $\Delta T_{2,1}$, show deviations from the predictions of GR, reaching up to 0.7674 min for Sgr A* and 1159.9 min for M87*. EHT observations place constraints on the EQG parameter ζ , with bounds $0 \leq \zeta \leq 0.9M$ for M87* and $0 \leq \zeta \leq 4M$ for Sgr A*, indicating that the constraint for M87* is tighter.

In conclusion, our analysis shows that gravitational lensing is a promising tool for testing quantum corrections in EQG theory. The constraints derived from the EHT data suggest that EQG black holes are consistent with current observations within a finite range of ζ . With future high-precision observations, particularly from the forthcoming ngEHT, these constraints could be further refined, potentially dis-

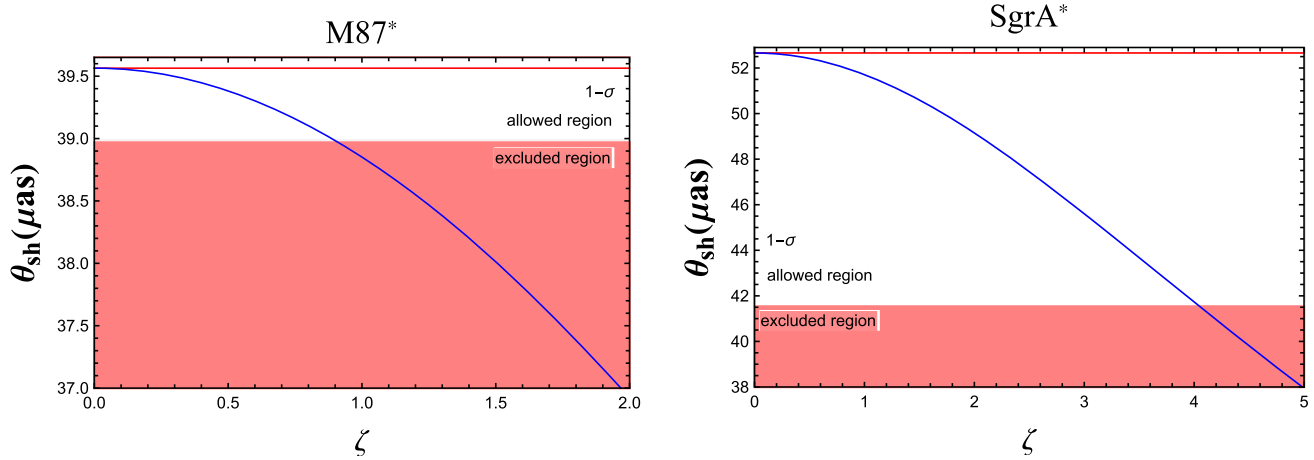


Fig. 9 Shadow angular diameter $\theta_{\text{sh}} (= 2\theta_{\infty})$ as a function of ψ , when M87* and Sgr A* are modelled as EQG black holes. The blue and red lines correspond to calculated values of $2\theta_{\infty}$ as a function of ζ for

Model-1 and Model-2, respectively. The white/red regions represent areas $1-\sigma$ consistent/inconsistent with the EHT observations, indicating that the latter imposed constraints on parameter ζ

tinguishing EQG black holes from classical Schwarzschild solutions. This approach offers an exciting avenue to deepen our understanding of black hole physics and quantum gravity.

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Appendix A: Deflection angle in the strong field limit

On solving the metric (3) on the equatorial plane ($\theta = \pi/2$) under null geodesics condition, we get,

$$\frac{d\phi}{dr} = \sqrt{\frac{Ar_0^2}{Br^2(A_0r^2 - Ar_0^2)}}, \quad (\text{A.1})$$

where the functions with subscript 0 are evaluated at r_0 . Then the deflection angle reads as

$$I(r_0) = 2 \int_{x_0}^{\infty} \frac{d\phi}{dr} dr, \quad (\text{A.2})$$

$$\alpha_D(r_0) = I(r_0) - \pi. \quad (\text{A.3})$$

For all spherically symmetric metrics, the deflection angle should be logarithmic [75], which can be rewritten in terms of impact parameter u as

$$\alpha_D(u) = -\bar{a} \log\left(\frac{u}{u_m} - 1\right) + \bar{b} + \mathcal{O}(u - u_m), \quad (\text{A.4})$$

with $u_m = u(r_0 = r_m)$, and $\bar{a}$ and $\bar{b}$ denoting the lensing coefficients. We define a new variable

$$z = \frac{A - A_0}{1 - A_0} \quad (\text{A.5})$$

where A is a function of r , and r_0 is the closest approach distance. Then, the integral (A.2) in the deflection angle becomes

$$I(r_0) = \int_0^1 R(z, r_0) F(z, r_0) dz, \quad (\text{A.6})$$

Table 7 Magnifications of first-order and second-order relativistic images due to lensing by Sgr A* with $d = D_{LS}/D_{OS} = 1/2$: EQG Model-2, which is equivalent to the Schwarzschild black hole and EQG Model-1 ($\zeta = 2M$) predictions for magnifications μ_n is given for different values of angular source position ψ . (a) $1p$ and $1s$ refer to first-order relativistic images on the same side as primary and secondary images, respectively. (b) We have used $M_{Sgr A^*} = 4.3 \times 10^6 M_\odot$, $D_{OL} = 8.3 \times 10^6$ pc

ψ	EQG					GR				
	$\mu_{1p, EQG}$	$\mu_{2p, EQG}$	$\mu_{1s, EQG}$	$\mu_{2s, EQG}$		$\mu_{1p, GR}$	$\mu_{2p, GR}$	$\mu_{1s, GR}$	$\mu_{2s, GR}$	
10^0	8.73281×10^{-13}	2.53415×10^{-16}	-8.73281×10^{-13}	-2.53415×10^{-16}		8.52495×10^{-12}	1.59×10^{-14}	-8.52495×10^{-12}	-1.59×10^{-14}	
10^1	8.73281×10^{-14}	2.53415×10^{-17}	-8.73281×10^{-14}	-2.53415×10^{-17}		8.52495×10^{-13}	1.59×10^{-15}	-8.52495×10^{-13}	-1.59×10^{-15}	
10^2	8.73281×10^{-15}	2.53415×10^{-18}	-8.73281×10^{-15}	-2.53415×10^{-18}		8.52495×10^{-14}	1.59×10^{-16}	-8.52495×10^{-14}	-1.59×10^{-16}	
10^3	8.73281×10^{-16}	2.53415×10^{-19}	-8.73281×10^{-16}	-2.53415×10^{-19}		8.52495×10^{-15}	1.59×10^{-17}	-8.52495×10^{-15}	-1.59×10^{-17}	
10^4	8.73281×10^{-17}	2.53415×10^{-20}	-8.73281×10^{-17}	-2.53415×10^{-20}		8.52495×10^{-16}	1.59×10^{-18}	-8.52495×10^{-16}	-1.59×10^{-18}	

Table 8 Magnifications of first and second-order relativistic images due to lensing by M87* with $d = D_{LS}/D_{OS} = 1/2$: EQG Model-2 which is equivalent to the Schwarzschild black hole and EQG Model-1 ($\zeta = 2M$) predictions for magnifications μ_n are given for different values of angular source position ψ . (a) $1p$ and $1s$ refer to first-order relativistic images on the same side as primary and secondary images, respectively. (b) We have used, $M_{M87^*} = 6.5 \times 10^9 M_\odot$, $D_{OL} = 16.8 \times 10^6$ pc

ψ	EQG					GR				
	$\mu_{1p, EQG}$	$\mu_{2p, EQG}$	$\mu_{1s, EQG}$	$\mu_{2s, EQG}$		$\mu_{1p, GR}$	$\mu_{2p, GR}$	$\mu_{1s, GR}$	$\mu_{2s, GR}$	
10^0	4.92945×10^{-13}	1.43046×10^{-16}	-4.92945×10^{-13}	-1.43046×10^{-16}		4.75466×10^{-12}	8.86797×10^{-15}	-4.75466×10^{-12}	-8.86797×10^{-15}	
10^1	4.92945×10^{-14}	1.43046×10^{-17}	-4.92945×10^{-14}	-1.43046×10^{-17}		4.75466×10^{-13}	8.86797×10^{-16}	-4.75466×10^{-13}	-8.86797×10^{-16}	
10^2	4.92945×10^{-15}	1.43046×10^{-18}	-4.92945×10^{-15}	-1.43046×10^{-18}		4.75466×10^{-14}	8.86797×10^{-17}	-4.75466×10^{-14}	-8.86797×10^{-17}	
10^3	4.92945×10^{-16}	1.43046×10^{-19}	-4.92945×10^{-16}	-1.43046×10^{-19}		4.75466×10^{-15}	8.86797×10^{-18}	-4.75466×10^{-15}	-8.86797×10^{-18}	
10^4	4.92945×10^{-17}	1.43046×10^{-20}	-4.92945×10^{-17}	-1.43046×10^{-20}		4.75466×10^{-16}	8.86797×10^{-19}	-4.75466×10^{-16}	-8.86797×10^{-19}	

where

$$R(z, r_0) = \frac{2(1 - A_0)\sqrt{Ar_0^2}}{r^2 A' \sqrt{B}}, \quad (\text{A.7})$$

$$F(z, r_0) = \frac{1}{\sqrt{A_0 - [(1 - A_0)z + A_0] \frac{r_0^2}{r^2}}}. \quad (\text{A.8})$$

The function $R(z, r_0)$ is regular for all z , but $F(z, r_0)$ will diverge as $z \rightarrow 0$. To avoid the divergence, we expand the $F(z, r_0)$ to the second order in z using a Taylor series. We define the square root term of $F(z, r_0)$ as a function $\Delta(z)$:

$$\Delta(z) = A_0 - [(1 - A_0)z + A_0] \frac{r_0^2}{r^2}. \quad (\text{A.9})$$

Expanding the $\Delta(z)$ to the second order in z , we obtain:

$$\Delta(z) \simeq \Delta(0) + \frac{z}{1!} \Delta'(0) + \frac{z^2}{2!} \Delta''(0) + \mathcal{O}(z^3). \quad (\text{A.10})$$

Since $\Delta(0) = 0$, we get

$$\bar{F}(z, r_0) \simeq \frac{1}{\sqrt{\Delta(z)}} \equiv \frac{1}{\sqrt{\frac{\Delta'(0)}{1!}z + \frac{\Delta''(0)}{2!}z^2}} \equiv \frac{1}{\sqrt{c_0 z + d_0 z^2}}. \quad (\text{A.11})$$

where,

$$c_0 = \frac{1}{1!} \frac{\partial}{\partial z} \left[A_0 - [(1 - A_0)z + A_0] \frac{r_0^2}{r^2} \right] \Big|_{r=r_0} \\ = \frac{1 - A_0}{r_0^2 A'_0} (2r_0 A_0 - r_0^2 A'_0) \quad (\text{A.12})$$

$$d_0 = \frac{1}{2!} \frac{\partial^2}{\partial z^2} \left[A_0 - [(1 - A_0)z + A_0] \frac{r_0^2}{r^2} \right] \Big|_{r=r_0} \\ = \frac{(1 - A_0)^2}{2r_0^4 A_0'^3} [4r_0^3 A_0'^2 + 2(r_0^2 - 4r_0^2) A_0 A'_0 - 2r_0^3 A_0 A_0'']. \quad (\text{A.13})$$

To solve Eq. (A.6), we split it into two parts: I_D which captures the divergent part, and I_R which is regular and obtained by subtracting the divergence from the original integral. In other words, $I(r_0) = I_D(r_0) + I_R(r_0)$ with

$$I_D(r_0) = \int_0^1 R(0, r_m) \bar{F}(z, r_0) dz, \quad (\text{A.14})$$

$$I_R(r_0) = \int_0^1 [R(z, r_0) F(z, r_0) - R(0, r_m) \bar{F}(z, r_0)] dz. \quad (\text{A.15})$$

The integral $I_D(r_0)$ can be solved exactly, yielding

$$I_D(r_0) = R(0, r_m) \frac{2}{\sqrt{d_m}} \log \frac{\sqrt{d_m} + \sqrt{c_m + d_m}}{\sqrt{c_m}}, \quad (\text{A.16})$$

where,

$$c_m = \frac{2d_m A'_m}{1 - A_m} (r_0 - r_m) + \mathcal{O}(r_0 - r_m)^2, \quad (\text{A.17})$$

and

$$d_m = d_0|_{r_0=r_m} = \frac{r_m^2 (1 - A_m)^2 (2A_m - A_m'' r_m^2)}{4A_m^2 r_m^2}. \quad (\text{A.18})$$

Rearranging all terms, we find

$$I_D(r_0) = -a \log \left(\frac{r_0}{r_m} - 1 \right) + b + \mathcal{O}(r_0 - r_m), \quad (\text{A.19})$$

where

$$a = \frac{R(0, r_m)}{\sqrt{d_m}} \quad (\text{A.20})$$

$$b = a \log \frac{2(1 - A_m)}{A'_m r_m}. \quad (\text{A.21})$$

Expanding impact parameter u in terms of critical impact parameter u_m at $r_0 = r_m$, we get

$$u - u_m = c (r_0 - r_m)^2 \quad (\text{A.22})$$

$$u_m \left(\frac{u}{u_m} - 1 \right) = c r_m^2 \left(\frac{r_0}{r_m} - 1 \right)^2 \quad (\text{A.23})$$

$$\sqrt{\frac{u_m}{c r_m^2} \left(\frac{u}{u_m} - 1 \right)} = \left(\frac{r_0}{r_m} - 1 \right), \quad (\text{A.24})$$

where $u'(r_m) = 0$ and

$$c = u''(r_m)/2! = \frac{2A_m - r_m^2 A_m''}{4\sqrt{A_m^3 r_m^2}} \\ = d_m \sqrt{\frac{A_m}{r_m^6} \frac{r_m^2}{(1 - A_m)^2}}. \quad (\text{A.25})$$

Now, plugging the above values in Eq. (A.19), we get

$$I_D(r_0) = -a \log \sqrt{\frac{u_m}{c r_m^2} \left(\frac{u}{u_m} - 1 \right)} + b + \mathcal{O}(r_0 - r_m), \quad (\text{A.26})$$

$$I_D(r_0) = -\frac{a}{2} \log \left(\frac{u}{u_m} - 1 \right) \\ + b - \frac{a}{2} \log \sqrt{\frac{u_m}{c r_m^2}} + \mathcal{O}(r_0 - r_m), \quad (\text{A.27})$$

For $I_R(r_m)$, the regular integral is computed numerically at $r_0 = r_m$. From Eq. (A.3), the deflection angle α_D as a function of the impact parameter u is expressed as [75]

$$\alpha_D(u) = -\bar{a} \log \left(\frac{u}{u_m} - 1 \right) + \bar{b} + \mathcal{O}(u - u_m), \quad (\text{A.28})$$

where, the lensing coefficients $\bar{a}$ and $\bar{b}$ are defined as

$$\bar{a} = \frac{R(0, r_m)}{2\sqrt{d_m}}, \quad (\text{A.29})$$

$$\bar{b} = -\pi + I_R(r_m) + \bar{a} \log \left(\frac{2d_m}{A_m} \right). \quad (\text{A.30})$$

Appendix B: EHT-imaging (Event Horizon Telescope imaging algorithm)

The Event Horizon Telescope (EHT) employs three imaging algorithms—EHT-Imaging, SMILI, and DIFMAP.

These independently reconstruct black hole images using the same dataset to minimize collective bias and enable comparative analysis. Each algorithm explores 10^3 – 10^5 parameter combinations, including weights for regularization functions, mask sizes, station gain constraints, and noise budgets. All three integrate low- and high-band data without flagging but differ in their initial models and exploration strategies [98].

B.1 EHT-imaging (Event Horizon Telescope imaging algorithm)

EHT-Imaging reconstructs high-resolution black hole images from VLBI data.

Purpose: Produces high-fidelity images of black hole shadows and photon rings, e.g., the M87 black hole.

Methodology:

Visibility Data: Processes complex Fourier analysis from Earth-based radio telescopes.

Image Reconstruction: EHT-Imaging uses regularized maximum likelihood (RML) methods, combining prior knowledge with regularization techniques like MEM and CLEAN for producing smooth images.

Calibration: Refinement of a simple geometrical initial model, and optimization of hyperparameters.

Regularization: To prevent overfitting, techniques like total variation (TV) minimization are applied.

Final Image: High-resolution images of the black hole's shadow, revealing detailed features such as photon rings.

Key Techniques: Maximum Likelihood Estimation (MLE), Bayesian Inference, and Regularization.

Applications: Imaging black hole shadows (e.g., M87* and Sgr A*) and accretion disk structures.

B.2 SMILI (sparse modeling imaging library for interferometry)

SMILI enhances image reconstruction accuracy using statistical methods, particularly with sparse interferometric data.

Purpose: Improves reconstruction accuracy by incorporating statistical models by focusing on sparse data scenarios.

Methodology:

Initial Model: Starts with basic structures like a point source or Gaussian disk and refines them iteratively via gradient descent.

Bayesian Framework: Maximizes posterior probability by integrating noise characteristics of data with the model assumptions.

Data Fitting: Adjusts parameters to the best-fit observed data while preserving consistency with the prior information.

Regularization: Added constraints (e.g. smoothness or sparsity) to avoid overfitting and ensure reconstructed features, like ring morphology.

Iterative Refinement: Uses methods like Markov Chain Monte Carlo (MCMC) to refine images and estimate uncertainties in the model.

Key Techniques: Bayesian Inference, MCMC, and Regularization.

Applications: High-resolution imaging, artifact reduction with low signal-to-noise ratio, and imaging of black hole accretion disks and jets.

B.3 DIFMAP (Difmap imaging algorithm)

DIFMAP specializes in imaging, calibration, and analysis of VLBI data.

Purpose: Used for reconstructing and data analysis of VLBI observations, focusing on compact sources such as quasars and black holes.

Methodology:

Visibility Data Processing: Calibrates phase and amplitude from complex visibility data.

CLEAN Deconvolution: The algorithm subtracts iteratively components from raw observed data, and hence creates a cleaner image of the source.

Model Fitting: Model-fitting techniques in which Gaussian components are fitted to observed data, separating different components of observed sources like shadows and accretion disks in black holes.

Imaging Algorithms: Employs various imaging algorithms such as Multi-Frequency Synthesis (MFS) for wide-band VLBI data.

Self-Calibration: Enhances image quality for low-SNR data by iteratively correcting phase and amplitude.

Key Techniques: CLEAN, Model Fitting, and Self-Calibration.

Applications: Imaging black hole accretion disks, jets, and detailed structures of radio sources.

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